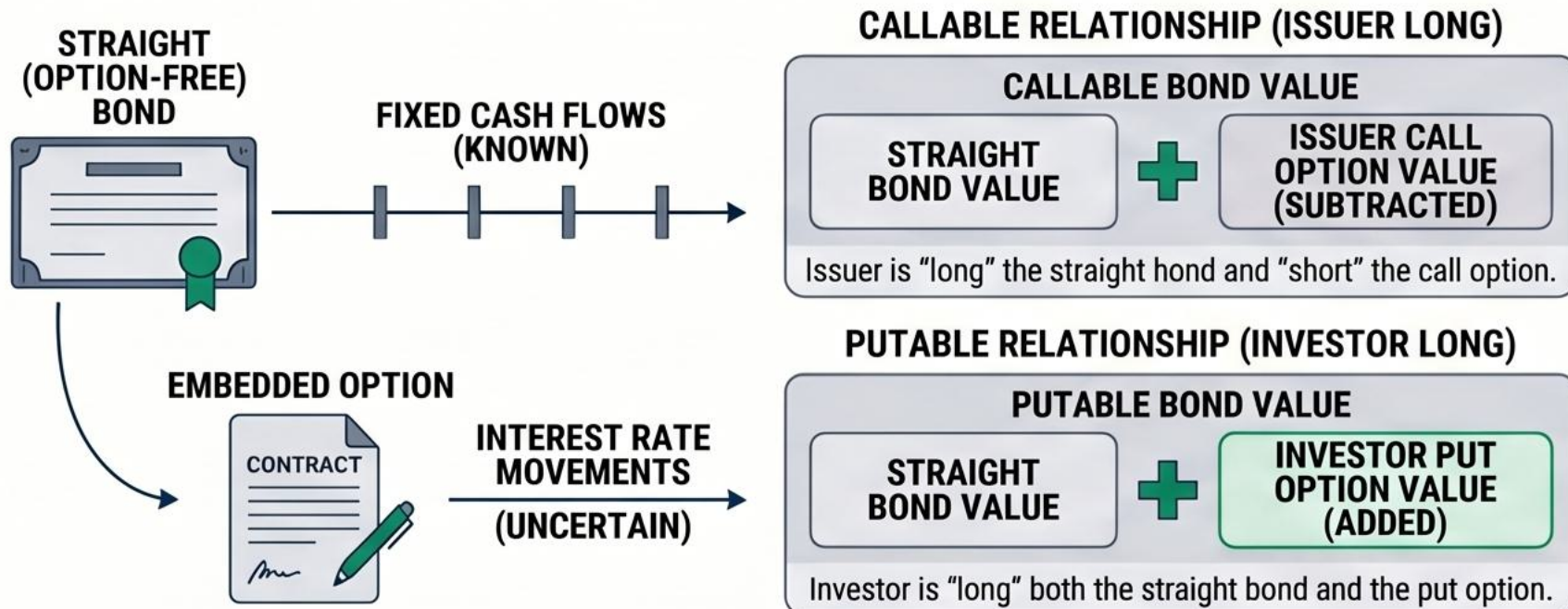


CFA Level II Mastery: Valuation & Analysis of Bonds with Embedded Options

THE BUILDING BLOCKS – STRUCTURAL RELATIONSHIPS



STEP-BY-STEP BINOMIAL TREE VALUATION

1. GENERATE INTEREST RATE TREE (FORWARD RATES & VOLATILITY)

$$\text{Callable Value} = \min(\text{Undiscounted Value}, \text{Call Price})$$

2. Determine Node Decisions

The determine node erotitance from the indpendent bonds roiders the value as ifthe aeration of the bonds connot of the two period set opts.

$$\text{Putable Value} = \max(\text{Undiscounted Value}, \text{Put Price})$$

Arbitrage-Free Framework: Value at each node is the average of the discounted values of the two possible next-period nodes.

3. BACKWARD INDUCTION (DISCOUNTING)

MEASURING RISK – DURATION AND CONVEXITY

$$EffDur = \frac{PV_- - PV_+}{2 \cdot \Delta curve \cdot PV_0}$$

Effective Duration (EffDur): The only appropriate duration measure for bonds with embedded options.

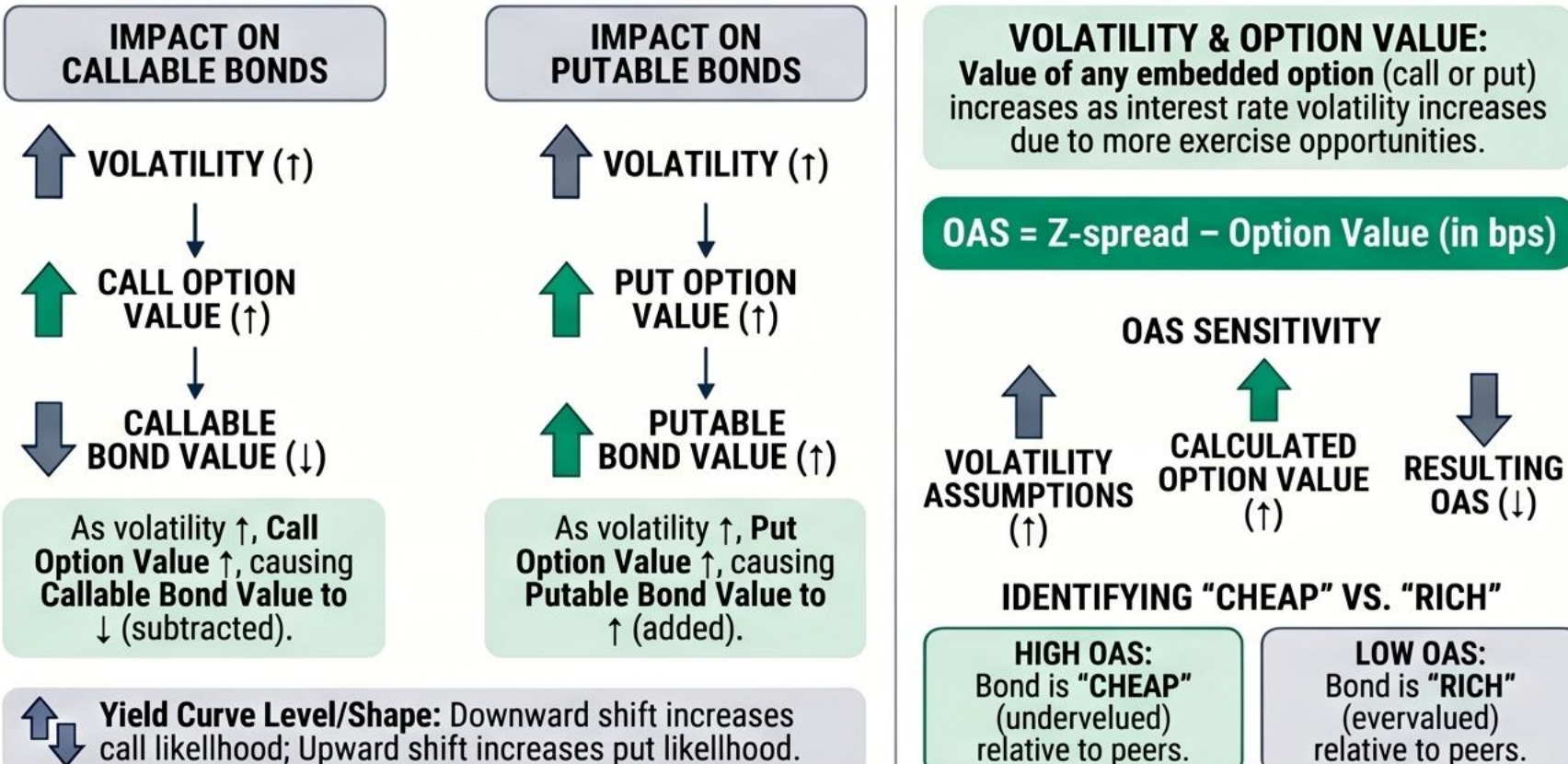
EFFECTIVE CONVEXITY

STRAIGHT BONDS:	Always positive.
CALLABLE BONDS:	Negative convexity when "near the money" (low yields); call price caps appreciation.
PUTABLE BONDS:	Always positive, appreciation enhanced as put price floors value.

Ona-Sided Durations: Used for axymmetrical price sensitivity. Callable (up-duration + down-duration); Putable (more sensitive to rate decreases).

Key Rate Duration: Measures sensitivity to specific yield curve point shifts (e.g., 10-year par bond only affected by 10-year key rate duration).

THE IMPACT OF INTEREST RATE VOLATILITY



COMPLEX VARIATIONS & EXAM TRAPS

EXAMPLE: CONVERTIBLE BONDS

$$\text{Value} = \text{Straight Bond Value} + \text{Value of Call Option on Stock}$$

If also callable: Value = Straight Bond + Stock Call - Issuer Call Option.

CAPS & FLOORS

CAPPED FLOATER: Straight Bond + Short Cap (benefits issuer).

FLOORED FLOATER: Straight Bond + Long Floor (benefits investor).

EXAM TRAPS

- THE "YIELD-TO-WORST" TRAP:** Candidates forget that for callable bonds, the lowest of all possible Yield-to-Calls and the YTM is the "Yield-to-Worst."
- THE BINOMIAL TREE TRAP:** In exam problems, always check the "Lockout Period." Options cannot be exercised during this time, so node values are not capped/floored until the lockout expires.